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Improvement of the fatigue behaviour of Al 6061/20% SiC_p composites by means of titanium coatings

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Abstract

The present work shows that it is possible to increase the fatigue life and endurance limit of Al 6061/20% SiC_p composite by means of coatings of titanium sputtered at room temperature. Coatings seal defects always present on the external surface of the composite, retard crack growth thus lead to a remarkable improvement of fatigue behaviour. The effect of coating thickness on fatigue is discussed with reference to surface state, which has been investigated by scanning tunnelling microscopy (STM).

Keywords: A. Metal-matrix composites; A. Coating; B. Fatigue; D. SEM; Sputtering

1. Introduction

In aerospace and the transport industry, aluminium-matrix composites, containing typically 15–20% of discontinuous reinforcement in the form of short ceramic fibres or particles, have attracted increasing interest because of their improved mechanical properties as compared with those of the corresponding monolithic alloys. Several 6xxx Al alloys, reinforced with SiC particles (SiC_p) or whiskers, show superior room-temperature fatigue life and endurance limit in high-cycle fatigue [1] but not in low-cycle fatigue [2]. At temperatures higher than 200 °C, the fatigue behaviour of composites is also better in low-cycle fatigue [3].

In these composites, fatigue life critically depends on crack initiation. In particular, the initiation site and subsequent short crack growth are affected by matrix microstructure and bonding conditions between particle and matrix, which are determined by the size and distribution of particles, chemistry [4] and processing routes [5]. Moreover, stress amplitude influences the crack density and crack-size distribution [6].

The dominant role is always played by the sample surface that represents a preferential site for crack nucleation. In spite of careful mechanical polishing, the surfaces of composites always exhibit a significant number of defects (voids, short and long cracks, fractured particles) connected to the specific material structure and an inhomogeneous stress distribution induced by the presence of a ceramic phase together with one or more metallic phases [7–9]. Therefore, the improvement of fatigue behaviour of composites requires: (i) elimination or substantial reduction of surface defects, (ii) homogeneous compressive stresses on the surface. Both of these points are beneficial for increasing fatigue life but the first, (i), is decisive. Since surface treatments (shot-peening, grit- or sand-blasting, rolling, etc.), which are usually employed on metals, partially lose their efficiency when applied to composites, it seems interesting to test other techniques, in particular sputtering deposition, ion implantation, ion mixing and dynamic ion mixing [10]. Although many papers may be found in the literature about the application of these techniques on metallic alloys, very little work has been done with composites.

In this work we have investigated the possibility of increasing the fatigue life of Al 6061/20% SiC_p by means of thin titanium coatings obtained by sputtering. The scope of coating is to improve the state of the surface by sealing its defects. A sputtering technique has

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been chosen because it guarantees homogeneity of deposition and good adhesion to the substrate. Deposition treatments have been carried out at room temperature to avoid microstructural modifications in the bulk. Titanium has good mechanical properties and very low diffusivity in aluminium up to high temperature, so it is possible to evaluate the sealing effect by itself without spurious effects due to interdiffusion and/or chemical reactions between atomic species of coating and substrate. The effect of coating thickness has been studied too.

2. Experimental

The Al 6061/SiC_p composite examined was produced by powder metallurgy, then extruded at 470 °C in the form of bars with a diameter of 20 mm. The nominal compositions of matrix and reinforcement are reported in Tables 1 and 2, respectively. SiC particles have a bimodal size distribution with peaks at 2 and 10 μm. Their concentration is 20% by volume.

Fatigue probes are 120 mm long with a diameter of 16 mm, the test zone is 7 mm long with a cross-sectional diameter of 7.4 mm. The test zone was mechanically polished to reduce defects due to probe preparation. In spite of that, the external surfaces still exhibited short cracks (< 100 μm), microcavities and voids caused by the detachment of particles during preparation, fractured particles and long cracks (> 100 μm), associated to voids and fractured particles (Fig. 1).

The experimental set-up used for coating fatigue probes with titanium by sputtering is schematically shown in Fig. 2. After mechanical polishing and degreasing three groups of probes were coated in the following conditions: power density on the titanium target ~4 W/cm^2, argon as working gas at a pressure of 0.5 Pa, temperature 30 °C, deposition rate 0.1 nm/s. The coating thickness of the three groups of probes was 1.0, 1.5 and 2.0 μm.

Both coated and uncoated probes were tested using a rotating bending machine with a single-end cantilever ($R = -1$, frequency = 17 Hz).

Scanning electron microscopy (SEM) observations were performed on the lateral surface of the test zone

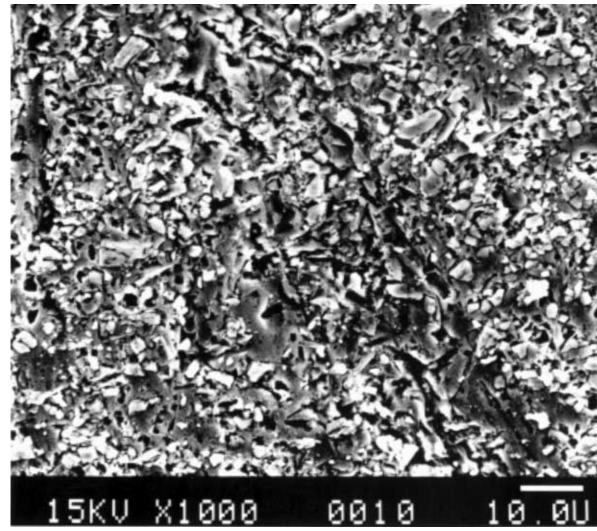

Fig. 1. External surface of an uncoated fatigue probe after mechanical polishing.

(before testing and at successive steps of fatigue life) and on the fracture surface.

Three groups of flat samples with the same coating thicknesses of fatigue probes were prepared for examining the surface by scanning tunnel microscopy (STM) and for investigating by X-ray diffractometry (XRD) the residual stresses induced in the matrix by coating. The main effect of residual stresses is to change mean interplanar spacings to shift peak positions. XRD spectra of coated and uncoated samples have been recorded in the 2Θ range 50–160° by step-scanning with steps of 0.005° and counting time of 10 s per step. To get the best precision in determining the angular position of peaks, the measurements have been carried out using Cr-K_α radiation ($\lambda = 0.22897$ nm).

Flat samples were also used to determine the Ti-film intrinsic hardness using the method of Jönsson and Hogmark [11], which requires conventional Vickers microhardness measurements with different loads on both coated and uncoated samples and the knowledge of coating thickness. On each specimen, indentations were made with eight loads ranging from 15 to 1000 g and 30 tests were made for each load.

Table 1
Nominal composition of the matrix (wt.%)

Fe	Si	Mg	Cu	Cr	Mn	Ti	Al
0.16	0.75	1.01	0.30	0.18	0.002	0.003	To balance

Table 2
Nominal composition of the reinforcement particles (wt.%)

SiC	SiO_2	C	Fe	Al	Ti	Ca
99	0.25	0.51	0.03	0.17	0.02	0.02

3. Results

Wöhler curves obtained by testing coated and uncoated probes are plotted in Fig. 3. N_F is the number of cycles to fracture, σ_{MAX} the maximum amplitude of alternating stress and σ_e the endurance limit.

From data reported in Fig. 3, it is clear that Ti coatings improve fatigue life and endurance limit. For $\sigma_{MAX} = 141$ MPa, N_F of uncoated probes is ~10^5, whereas the value found for probes with 1.5-μm thick Ti coating is one

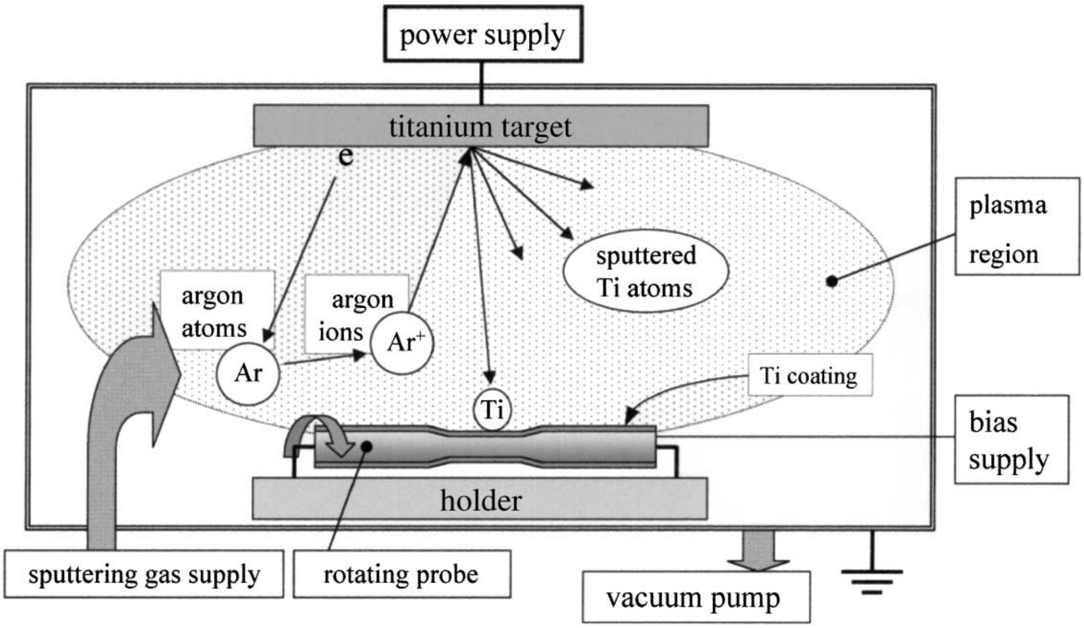

Fig. 2. Sketch of the experimental set-up used for coating the samples by Ti sputtering deposition. To achieve an homogeneous coating thickness, fatigue probes rotate under the beam.

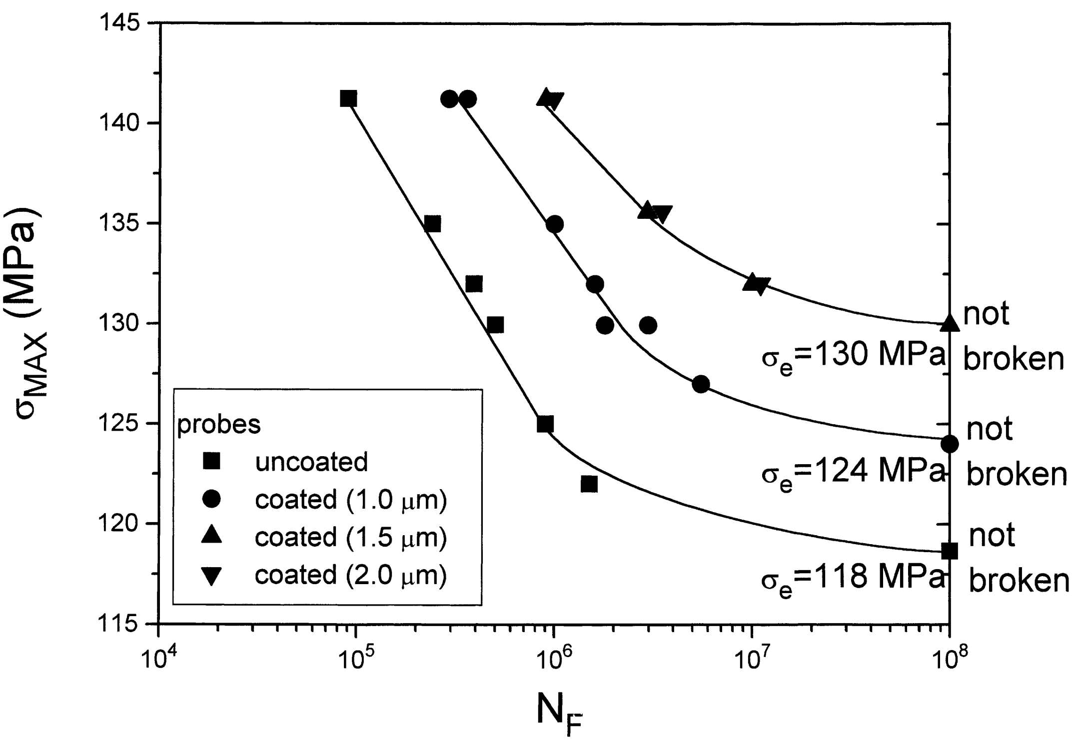

Fig. 3. Wöhler curves of coated and uncoated probes ($R = -1$).

order of magnitude higher ($N_F \sim 10^6$). The difference increases in the region of lower applied stresses. If one considers that fatigue life of the Al 6061/20% SiC_P composite is about 5 times that of the corresponding monolithic alloy [5], the remarkable effect due to Ti films is evident.

For a given σ_{MAX}, N_F increases as coating thickness increases up to 1.5 μm while further improvement is not observed with thicker coatings (2 μm). The trend of N_F versus coating thickness is reported in Fig. 4.

In addition, the endurance limit σ_e depends on coating thickness: its value, which is $\sim$118 MPa for uncoated probes, reaches $\sim$130 MPa for a Ti coating thickness of 1.5 μm (Fig. 3).

Ti films exhibit good adhesion to the probes and SEM observations at successive steps of fatigue life show that

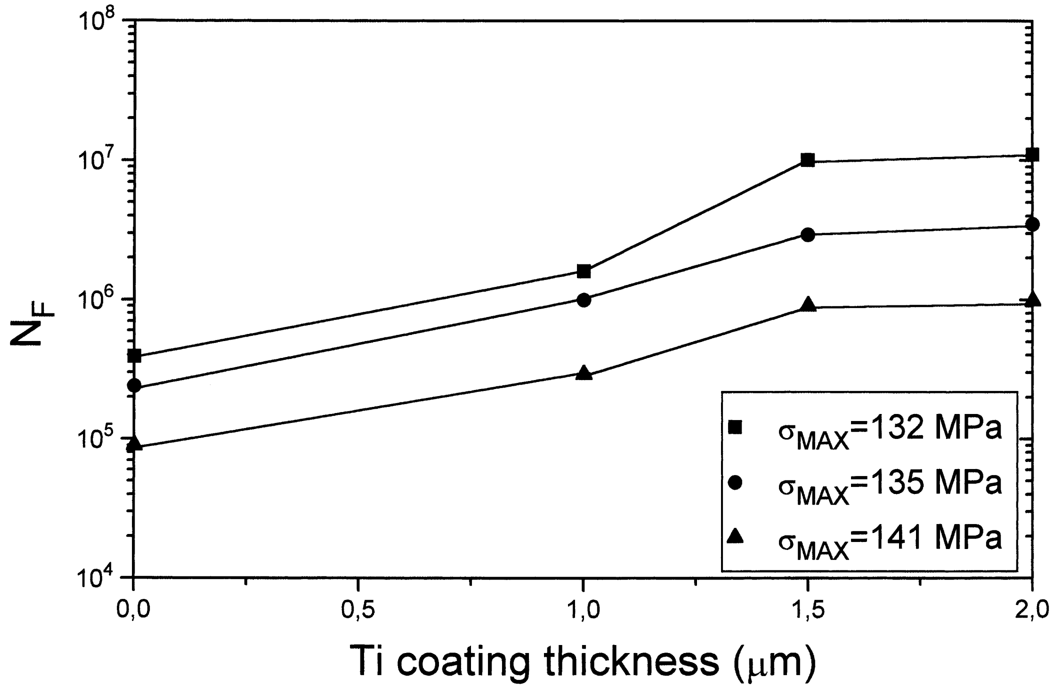

Fig. 4. The number of cycles to fracture, N_F, is plotted versus Ti coating thickness for three different σ_{MAX} values.

coatings maintain their integrity up to $\sim 0.9\ N_F$. In Fig. 5a, cracks and delaminations are present in the coating near the zone of fracture. In correspondence of crack initiation sites, protrusion of material is sometimes observed (Fig. 5b). Fracture surfaces of coated and uncoated probes have similar morphology, which has been described in detail in Ref. [9].

STM observations enabled us to evaluate the surface conditions with different coating thicknesses. Fig. 6 shows STM images of samples with (a) 1, (b) 1.5 and (c) 2 μm thick Ti coatings and profiles recorded along marker lines. Profiles in (a) and (b) exhibit peaks with different mean heights: (a) ~500 nm, (b) ~100 nm. When coating thickness is further increased (Fig. 6c), samples have surface conditions similar to those shown in (b) with mean peak heights of ~100 nm. The reported profiles indicate that roughness decreases as coating thickness increases up to 1.5 μm, then it remains nearly constant.

The hardness of the 2-μm thick Ti film was determined using the method of Jönsson and Hogmark [11]. Taking into account the geometry of Vickers imprint, the intrinsic hardness H_{F0} of the coating may be written as:

$$H_{F0} = H_{S0} + \frac{(k_C - k_S)}{2\chi t} \quad (1)$$

where H_{S0} is the intrinsic hardness of the substrate (Al 6061/20% SiC_p composite), χ a constant depending on the deformation mode of coating and t the coating thickness.

For a given penetration depth d of the imprint, k_C and k_S, constants characteristic of the system (film + substrate)

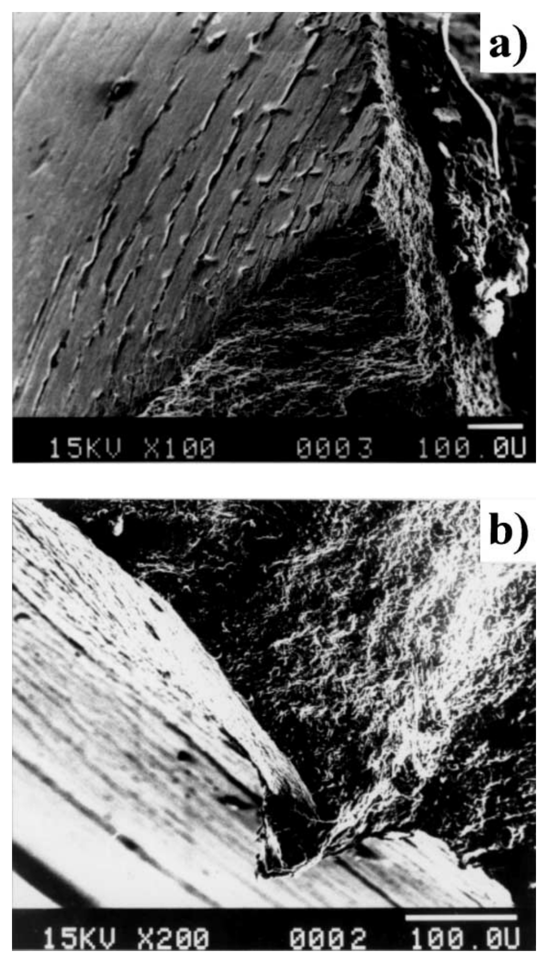

Fig. 5. (a) Cracks and delaminations in the coating near the zone of fracture; (b) Protrusion of material in correspondence of crack initiation site.

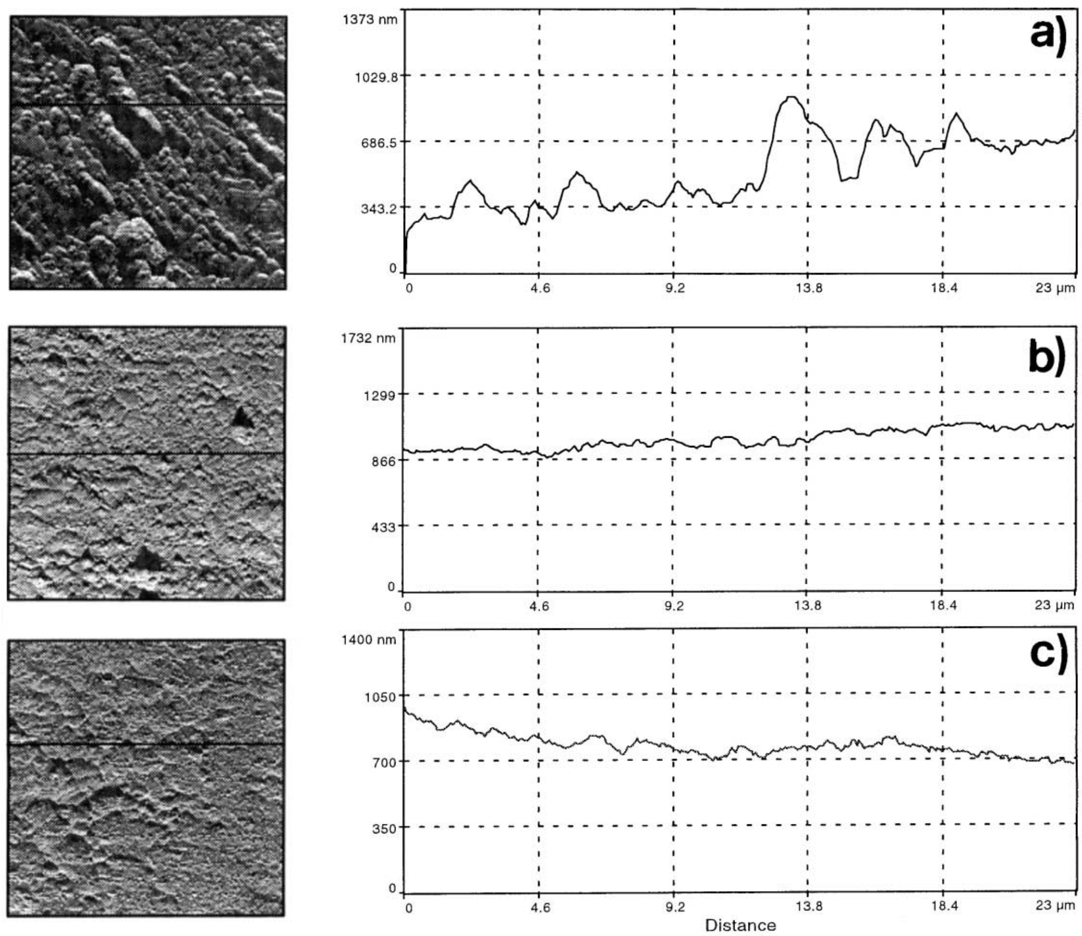

Fig. 6. STM images of samples coated with (a) 1μm, (b) 1.5 μm and (c) 2 μm of Ti. The reported profiles have been recorded along the marker lines.

and substrate, respectively, verify the following relationships:

$$H_C(d) = H_{C0} + \frac{k_C}{d} \quad (2)$$

$$H_S(d) = H_{S0} + \frac{k_S}{d} \quad (3)$$

where $H_C(d)$ and $H_S(d)$ mean hardness values measured with different loads on coated and uncoated samples, H_{C0} the intrinsic hardness of the system.

H_{C0} was determined by extrapolating to zero the plot $H_C(d)$ vs *1/d*. The slope of the plot gives k_C .

H_{S0} and k_S were determined in the same way from hardness data of samples without coating. In this case only hardness data obtained with 300, 500 and 1000 g were considered; for lower loads data scattering is quite large because microhardness measurements are affected by the local distribution of particles in the zone of indentation.

SEM observations show that Ti film cracks on the rim and inside the imprint (Fig. 7). Since the fracture mode of Ti sputtered coating is brittle, the value of χ was taken equal to 0.5 according to Ref. [11]. So, introducing in Eq. (1) the values $H_{S0} = 76$ kg/mm^2, $k_C = 1.47$ kg/mm, $k_S = 0.41$ kg/mm and $t = 0.002$ mm, it was calculated $H_{F0} = 606$ kg/mm^2.

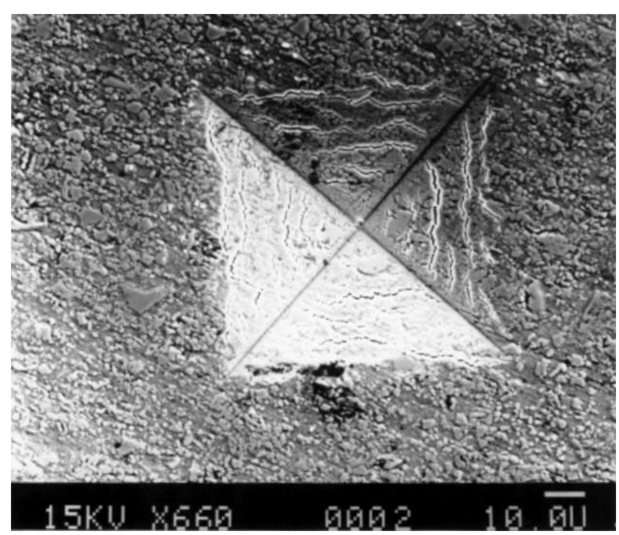

Fig. 7. Vickers imprint on a sample with a 2-micron thick Ti coating. Cracks are visible inside the indentation and on its rim.

A part of XRD spectra of coated and uncoated samples is shown in Fig. 8. Due to the strong texture of extruded material Al (220) and Al (311) peaks ($2\Theta = 106.3$ and $139.3°$, respectively) have a very low intensity, at the limit of detectability; thus it was not possible to perform precise measurements of their positions. As shown in Fig. 8, a similar difficulty was encountered in the case of Al (111) reflection of coated samples for its overlapping with the broad Ti (002) peak. Therefore, Al (200) peak is the only diffraction line which can give information about the possible presence of residual stresses in the matrix. The Al (200) peaks of uncoated and coated samples are centred at $2\Theta = 68.72°$ and at $2\Theta = 68.725°$, respectively; thus, the angular difference (0.005°) is within the experimental uncertainty.

4. Discussion

Fatigue life consists of both crack initiation and crack propagation. In a previous paper [8], crack propagation in this material has been described by means of SEM observations on uncoated samples. Crack paths avoid SiC particles in the first part of fatigue life whereas SiC particles of larger size are fractured in the last stage. Crack propagation diagrams (growth rate da/dN versus applied stress intensity ΔK) reported in literature (e.g. see Ref. [5]) for particle reinforced composites are sigmoidal in shape and are described by the Paris equation in the midrange of growth rates:

$$da/dN = C\Delta K^{m} \tag{4}$$

The constants C and m were experimentally determined for Al 6061/SiC_P in different fatigue conditions [12]. The Paris region is very short when compared with that of the corresponding monolithic alloy due to composite lower fracture toughness and relatively higher threshold value ΔK^* (if $\Delta K < \Delta K^*$ cracks are dormant).

During the first part of fatigue life (at low ΔK values) cracks tend to propagate by avoiding particles. This is ascribed to the tendency of the higher modulus ceramic phase to repel the crack [8,13,14]. On the contrary, at high ΔK values (reached in the reduced cross section of probe during the last part of fatigue life), crack growth accelerates, the size of the plastic zone ahead of the crack tip increases and fracture of the SiC particles in the vicinity becomes more likely.

The improved fatigue behaviour of composites with respect to their monolithic alloys is substantially a result of the particles which act as obstacles for crack propagation. However, the presence of reinforcement particles may be detrimental for crack initiation, as shown in Fig. 1, which shows a lot of surface defects associated with SiC particles.

With the aim of enhancing the fatigue life of the material, the samples were coated by sputtered Ti to eliminate possible crack-initiation sites. Ti films are considerably harder than the substrate. Coatings appear smooth without discontinuities and maintain integrity and good adhesion during a large part of the fatigue life.

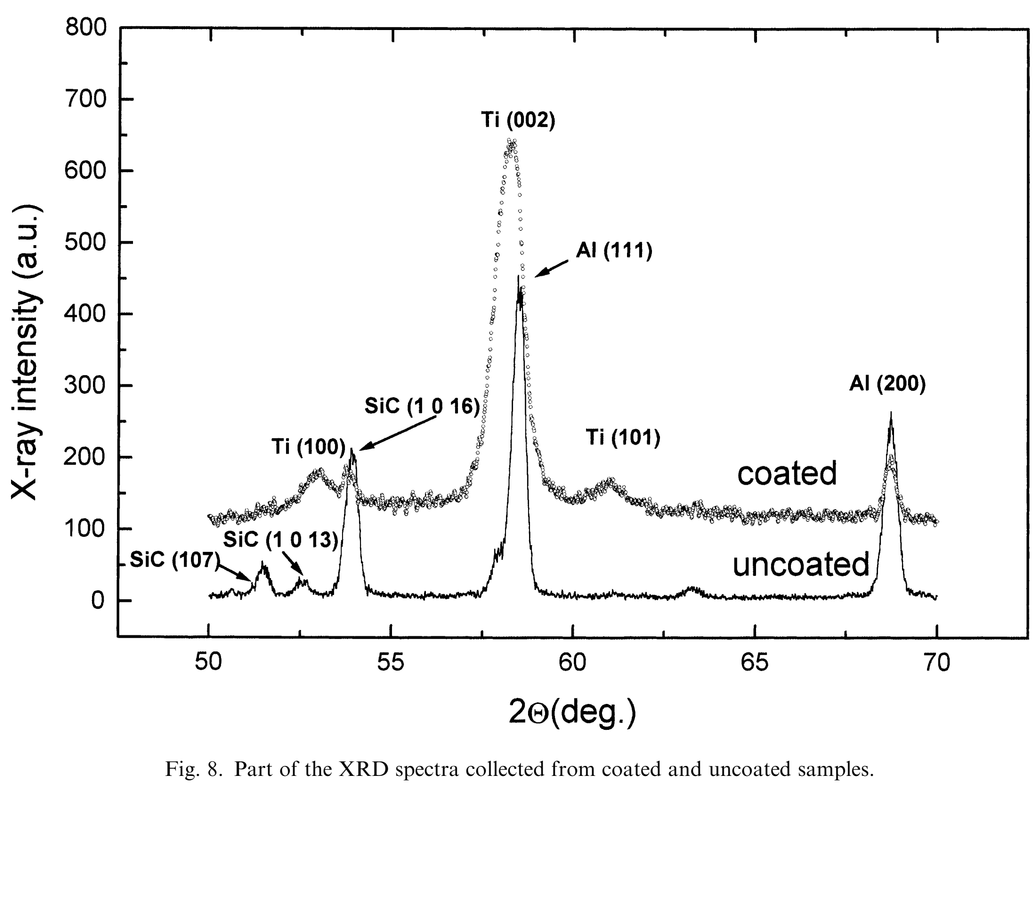

Fig. 8. Part of the XRD spectra collected from coated and uncoated samples.

Some defects are visible when the number of cycles exceeds 0.9 N_F and only after the fracture coating appears detached, particularly in correspondence of the zone where material protrusion is observed (Fig. 5a and b). Therefore, coatings guarantee a satisfactory sealing of surface defects.

The comparison of Wöhler curves (Fig. 3) of coated and uncoated probes clearly indicates the effectiveness of Ti coatings in improving fatigue behaviour with an increase of fatigue life and endurance limit.

STM observations show that surface roughness decreases as coating thickness increases up to 1.5 μm, in correspondence with this value, the best fatigue performances are obtained. The surface of Ti films thinner than 1.5 μm is affected by the corrugated morphology of the substrate. For coating thicknesses larger than 1.5 μm, the roughness remains substantially unchanged (Fig. 6) and no further improvement of fatigue life is observed. It is concluded that the improved fatigue life of coated probes is a result of better surface conditions determined by the given treatment.

To understand the phenomenon, it is worth considering the classical relationship [6] describing the dependence of ΔK on $\Delta\sigma$ for a single crack with initial depth a_0:

$$\Delta K = \Delta\sigma(\pi a_0)^{1/2} Y(a_0) \tag{5}$$

where $\Delta\sigma = \sigma_{MAX} - \sigma_{MIN}$ (being $R = -1$, $\Delta\sigma = 2\sigma_{MAX}$). $Y(a_0)$ is a geometry factor, which can be set equal to 1 supposing, for simplicity, that cracks are semi-elliptical.

According to Eq. (5), for a given $\Delta\sigma$ applied to the fatigue probe, a decrease of a_0 corresponds to a decrease of ΔK. The case for a composite is more complex because there are several surface defects with different size and shape, and in any case Ti coating decreases the depth of each of them retarding the start of crack motion.

Effects caused by residual stresses in the composite matrix have been considered too. However, Al (200) line positions of coated and uncoated samples do not show substantial differences, thus Ti coatings do not induce residual stresses in the metal matrix of substrate. Therefore, the contribution to fatigue-life enhancement from lattice strains in the substrate seems to be negligible.

Experiments on another composite (A359/20% SiCp), which has been tested also at the temperature of 200 °C, confirm the beneficial effect of Ti coatings on fatigue behaviour. The publication of these results is underway [15].

In the attempt to further improve the method, it is necessary to develop a stronger film anchorage to the substrate. The achievement could be obtained by combining sputtering deposition with ion implantation. Ion implantation can induce compressive stresses in the surfaces but it does not eliminate the presence of the defects acting as initiation sites for fatigue cracks. The deposition of a metallic film and the successive bombardment (ion mixing) or the deposition combined with a simoultaneous ion implantation (dynamic ion mixing [10]) seem promising to get stronger film/substrate bonds.

5. Conclusions

Fatigue behaviour of the Al 6061/20% SiC_p composite in rotating bending tests is strictly affected by the surface conditions of the probes. A relevant improvement of fatigue life and endurance limit was observed in probes previously coated by sputtered Ti. Ti films, which are considerably harder than the substrate, exhibit good adhesion for a large part of their fatigue life.

Fatigue life and endurance limit increase by increasing coating thickness until the surface roughness is no more affected by the morphology of the substrate. This condition is reached with a deposition thickness of 1.5 μm. Thicker coatings do not modify surface roughness and thus do not improve composite fatigue behaviour.

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